

Skills Summary

Degree–Radian and Quadrant–Sign Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Choosing the conversion factor

One fact, two uses: $180^\circ = \pi$ radians.

degrees $\rightarrow$ radians: multiply by $\frac{\pi}{180}$. radians $\rightarrow$ degrees: multiply by $\frac{180}{\pi}$.

Self-check: converting *from* radians, the π must cancel. A π still standing means the factor was upside down.

Example: Convert 135° to radians.

Step 1. Starting in degrees and wanting radians, so the factor is the one that puts π on top.

Step 2. $135 \cdot \frac{\pi}{180} = \frac{135\pi}{180} = \frac{3\pi}{4} \Rightarrow 135^\circ = \frac{3\pi}{4}$ radians.

Try it: Convert 300° to radians.

check: $\frac{\pi}{180}$

Skill 2 — Attaching the sign from the quadrant

ASTC (“All Students Take Calculus”): in quadrant I *all* are positive; in II only *sine*; in III only *tangent*; in IV only *cosine*.

Order of work: find the reference angle, read its value from the first quadrant, then attach the sign the quadrant demands. The quadrant never changes *which* ratio you read.

Example: Evaluate $\cos 225^\circ$ exactly.

Step 1. Want cosine of an angle past 180° , so find the reference angle first and let the quadrant supply the sign.

Step 2. $225 - 180 = 45$, so the reference angle is 45° and the angle is in quadrant III, where cosine is negative. $\Rightarrow \cos 225^\circ = -\cos 45^\circ = -\frac{\sqrt{2}}{2}$

Try it: Evaluate $\sin 300^\circ$ exactly.

check: $-\frac{\sqrt{3}}{2}$

(!) Watch out: an inverse-trig key returns one angle only — $\sin^{-1}$ and $\tan^{-1}$ from -90° to 90° , $\cos^{-1}$ from 0° to 180° . On a 360° interval there are usually *two* answers, and the calculator’s may not even be one of them.

Skill 3 — Reading the reference angle's value

The three you must know:

$$\sin 30^\circ = \frac{1}{2}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}; \quad \sin 45^\circ = \cos 45^\circ = \frac{\sqrt{2}}{2}; \quad \sin 60^\circ = \frac{\sqrt{3}}{2}, \quad \cos 60^\circ = \frac{1}{2}.$$

Tangent is the quotient: $\tan \theta = \frac{\sin \theta}{\cos \theta}$.

Example: Evaluate $\cos \frac{4\pi}{3}$ exactly.

Step 1. Want cosine, so I need the cosine of the reference angle — the sine value is the classic wrong pick here.

Step 2. $\frac{4\pi}{3} = \pi + \frac{\pi}{3}$: reference angle $\frac{\pi}{3}$, quadrant III, cosine negative, and $\cos \frac{\pi}{3} = \frac{1}{2}$.

$$\Rightarrow \cos \frac{4\pi}{3} = -\frac{1}{2}$$

Try it: Evaluate $\tan \frac{5\pi}{3}$ exactly.

check: $\frac{5\pi}{3}$